1 Upper Bounds on Decoding Performance

We estimate upper bounds on decoding performance based on (i) neural consistency across trials and (ii) the fraction of neural activity that is behavior-relevant under simplifying assumptions, including linear neural–behavior relations. Figure 1 summarizes these bounds and their relationship to empirical linear decoding performance.

1.1 Neural Consistency Upper Bound (Split-Half)

Notation. Let $y_{k,t} \in \mathbb{R}^N$ and $z_{k,t} \in \mathbb{R}^B$ denote neural activity and behavior, respectively, for trial $k \in \{1, \dots, K\}$ at time $t \in \{1, \dots, T\}$.

Split-half estimator. For each session, trials are randomly divided into two halves $\mathcal{K}_1, \mathcal{K}_2$, and we compute the corresponding PSTHs:

$$\bar{y}_t^{(1)} = \frac{1}{|\mathcal{K}_1|} \sum_{k \in \mathcal{K}_1} y_{k,t}, \tag{1}$$

$$\bar{y}_t^{(2)} = \frac{1}{|\mathcal{K}_2|} \sum_{k \in \mathcal{K}_2} y_{k,t}. \tag{2}$$

We compute the Pearson correlation r between $\bar{y}_t^{(1)}$ and $\bar{y}_t^{(2)}$, and apply the Spearman–Brown correction:

$$R^2_{\text{cap,neural}} = \frac{2r}{1+r}, \tag{3}$$

This provides a data-driven estimate of the fraction of neural variance that is consistent across trials.

Interpretation. As shown in Fig. 1, this bound is high for Musall (~ 0.95), reflecting strong trial alignment, and substantially lower for Kondo's successful trials (~ 0.34), reflecting weaker alignment and higher variability. Example PSTHs for both datasets are shown in Fig. 1, illustrating the difference in trial consistency.

While this analysis provides an upper bound on neural reliability, translating it into a bound on behavior decoding requires separating behavior-relevant and irrelevant components of neural signal, which we address next.

1.2 Behavior-Relevant Upper Bound

Signal decomposition and assumptions. We assume neural activity can be decomposed as:

$$y_{k,t} = s_t + n_{k,t}, \quad s_t = s_t^{\text{beh}} + s_t^{\text{irr}}, \tag{4}$$

where s_t^{beh} is behavior-relevant signal, s_t^{irr} is behavior-irrelevant signal, and $n_{k,t}$ is trial-specific noise.

We assume:

$$\text{Cov}(s^{\text{beh}}, s^{\text{irr}}) = 0, \tag{5}$$

$$\text{Cov}(s, n) = 0, \tag{6}$$

i.e., behavior-relevant signal, irrelevant signal, and noise are uncorrelated (a simplifying but commonly used assumption).

Behavior-relevant fraction. Let $\bar{y}_t$ and $\bar{z}_t$ denote neural and behavioral PSTHs. We estimate the behavior-relevant component via linear least squares:

$$\hat{s}_t^{\text{beh}} = \bar{z}_t W, \quad W = \arg\min_W \|\bar{y}_t - \bar{z}_t W\|^2. \tag{7}$$

We define:

$$\rho^2_{\text{beh}} = \frac{\text{Var}_t(\hat{s}_t^{\text{beh}})}{\text{Var}_t(\bar{y}_t)}, \tag{8}$$

which measures the fraction of neural variance explained by behavior.

Behavior-relevant bound. The corresponding bound can be written as:

$$R^2_{\text{cap,beh}} = \frac{\mathrm{Var}(s^{\text{beh}})}{\mathrm{Var}(y)} = \frac{\mathrm{Var}(s^{\text{beh}})}{\mathrm{Var}(s)} \cdot \frac{\mathrm{Var}(s)}{\mathrm{Var}(y)}. \tag{9}$$

The first term $\mathrm{Var}(s^{\text{beh}})/\mathrm{Var}(s)$ is estimated via ρ^2_{beh} and Eqs. 7 and 8, while the second term $\mathrm{Var}(s)/\mathrm{Var}(y)$ is estimated using the split-half neural bound $R^2_{\text{cap,neural}}$. Thus:

$$R^2_{\text{cap,beh}} = \rho^2_{\text{beh}} \cdot R^2_{\text{cap,neural}}. \tag{10}$$

Interpretation. As shown in Fig. 1, this yields approximately 0.53 (Musall) and 0.28 (Kondo). This bound is more conservative than the neural bound but relies on simplifying assumptions (e.g., linearity of neural-behavior relation, independence), and should therefore be interpreted as an informative approximation rather than a precise ceiling.

Across datasets, the empirical linear regression (LR) performance remains below these bounds, consistent with their interpretation as upper limits for linear neural-behavior relations (LR $< R^2_{\text{cap,beh}} < R^2_{\text{cap,neural}}$).

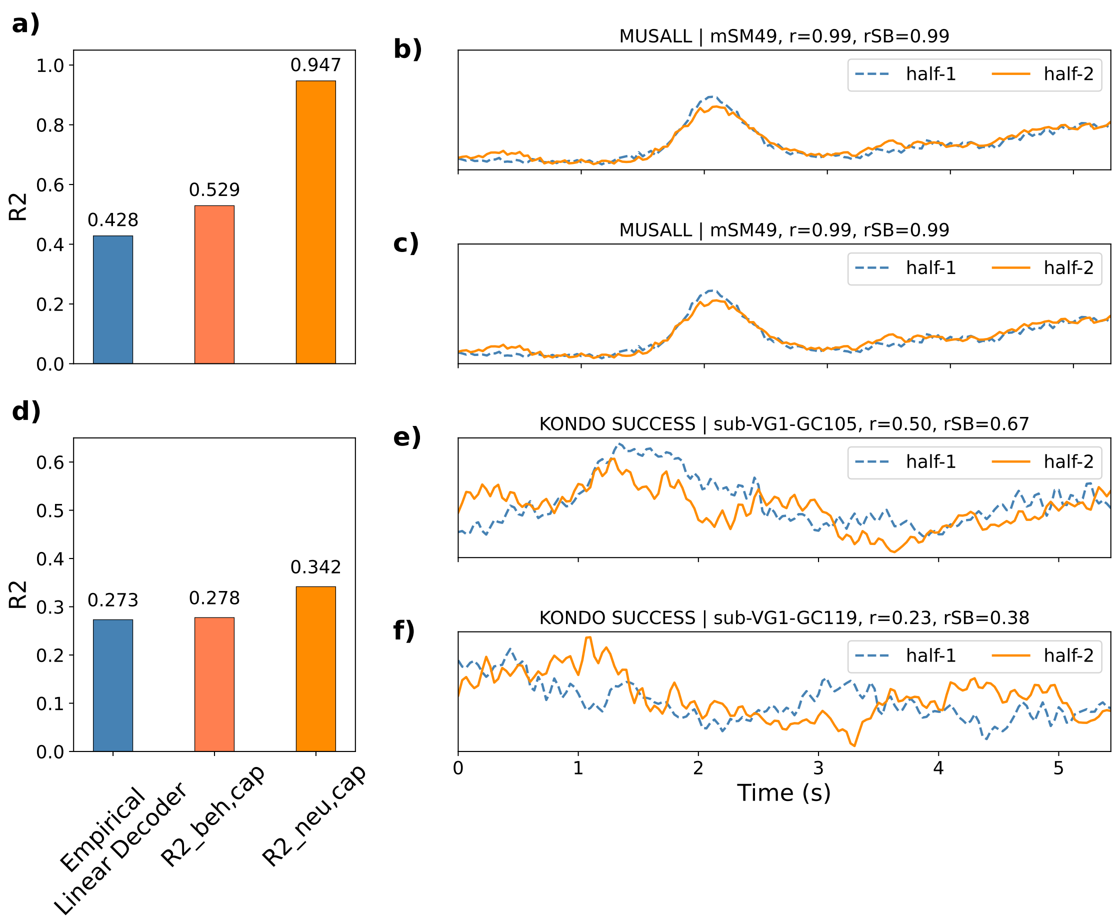

Figure 1: **Upper bounds on decoding performance and PSTH consistency across datasets.** **(a)** Musall dataset: comparison of empirical linear decoder performance, behavior-relevant bound ($R^2_{\text{cap,beh}}$; Sec. 1.2), and neural split-half bound ($R^2_{\text{cap,neural}}$; Sec. 1.1). The empirical performance lies below both bounds, consistent with their interpretation as upper limits for linear neural–behavior relations. **(b,c)** Example split-half PSTHs from two Musall sessions, showing strong agreement between halves and high correlation, indicating reliable trial alignment and stable bound estimation. For each pair, r denotes the correlation between the two halves, and $r_{\text{SB}} = \frac{2r}{1+r}$ is the Spearman–Brown corrected reliability. **(d)** Same analysis as (a) for the Kondo dataset. The empirical performance remains below the bounds, while the gap between bounds is reduced. **(e,f)** Example split-half PSTHs from Kondo sessions, showing weaker correspondence between halves due to weaker trial alignment and higher variability. The lower r and r_{SB} values illustrate that PSTH-based estimates are less reliable for Kondo, and the corresponding bounds should therefore be interpreted with caution compared to Musall.

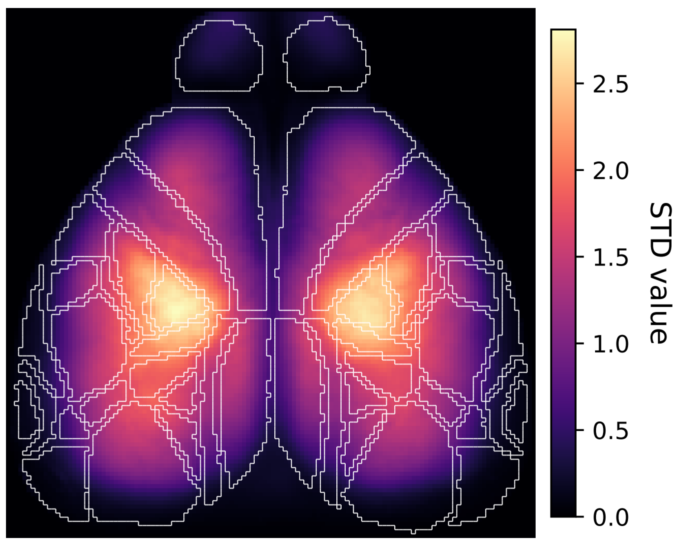

Figure 2: **Global standard deviation map.** Per-pixel standard deviation of widefield activity aggregated across all sessions. The Allen atlas boundaries are overlaid. Boundary regions consistently exhibit lower variance, indicating weaker neural signal and increased sensitivity to atlas misalignment in cross-subject settings.

Table 1: **Detailed subregion spatial variance statistics.** Rows report atlas-defined subregions grouped by their parent cortical area (group headers shown in bold). For each subregion, we report the spatial variance (Mean ± Std) across sessions and the number of pixels (N Pixels) belonging to that subregion.

ROI Group / Subregion	Spatial Variance	N Pixels
SS		
SSp_bfd_l	1.6588 ± 0.8758	321
SSp_bfd_r	1.5274 ± 0.8168	317
SSp_ll_l	6.7865 ± 0.6413	121
SSp_ll_r	6.4629 ± 0.4298	120
SSp_m_l	0.5249 ± 0.4549	139
SSp_m_r	0.5024 ± 0.4391	139
SSp_n_l	0.8501 ± 0.5742	95
SSp_n_r	0.7865 ± 0.5349	97
SSp_tr_l	5.3937 ± 0.7174	78
SSp_tr_r	5.2151 ± 0.7796	77
SSp_ul_l	4.0362 ± 1.1272	156
SSp_ul_r	3.7813 ± 1.1400	169
SSp_un_l	3.1185 ± 0.6549	44
SSp_un_r	2.8854 ± 0.6490	36
SSs_l	0.0992 ± 0.1124	112
SSs_r	0.0791 ± 0.0868	109
VIS		
VISa_l	3.5430 ± 0.3070	105
VISa_r	3.0627 ± 0.3569	105
VISal_l	0.5210 ± 0.2246	38
VISal_r	0.3704 ± 0.1538	40
VISam_l	2.5817 ± 0.3053	43
VISam_r	2.0931 ± 0.2757	50
VISl_l	0.0678 ± 0.1198	203
VISl_r	0.0387 ± 0.0766	204
VISp_l	1.0023 ± 0.9445	575
VISp_r	0.7482 ± 0.7785	586
VISpm_l	1.6791 ± 0.7971	70
VISpm_r	1.3272 ± 0.5487	65
VISrl_l	1.9793 ± 0.5846	57
VISrl_r	1.5836 ± 0.4908	56
MO		
MOp_l	1.9497 ± 1.2473	435
MOp_r	1.8810 ± 1.2545	434
MOs_l	0.9116 ± 0.6561	849
MOs_r	0.8374 ± 0.6019	846
AUD		
AUDp_l	0.0228 ± 0.0084	51
AUDp_r	0.0145 ± 0.0062	48
AUDs_l	0.0830 ± 0.0550	66
AUDs_r	0.0528 ± 0.0385	77
OB		
OB_l	0.0725 ± 0.0618	301
OB_r	0.0519 ± 0.0514	288
RSP		
RSPagl_l	1.5387 ± 1.2116	151
RSPagl_r	1.4316 ± 1.1503	154
RSPd_l	1.0950 ± 0.7201	376
RSPd_r	0.9480 ± 0.6823	367